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**Subject: Programming with Java**

# ASSIGNMENT No :4

**Subject: Programming with Java**

**TITLE :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.**

## Problem Statement :

**Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.**

## Problem Code:

EXP_3.java > ...

```java
public class EXP_3 {
    public static void main(String[] args) {
        // creating a one primitive variable.
        int marks1 = 88;
        System.out.println("Primitive Variable Value: " + marks1);

        // converting the primitive object into the wrapper class object.
        Integer obj1 = Integer.valueOf(marks1);
        System.out.println("Wrapper class object: " + obj1);
        // this process is called "BOXING".

        // AUTOBOXING.
        int marks2 = 70;
        Integer obj2 = marks2;
        System.out.println("Wrapper class object created using autoboxing: " + obj2);

        // Manual UNBOXING
        Integer obj3 = 33;
        int marks3 = obj3.intValue();
        System.out.println("Primitive datatype using manual unboxing: " + marks3);

        // automatic unboxing
        Integer obj4 = 45;
        int marks4 = obj4;
        System.out.println("Primitive datatype using automatic unboxing: " + marks4);

        // String to int conversion
        String str = "250";
        int num = Integer.parseInt(str);
        System.out.println("Number is: " + num);

        // conversion of a wrapper object to a string
        Integer marks5 = 67;
        String abc = marks5.toString();
        System.out.println("String is: " + abc);

        // arithmetic operations on a wrapper class object.
        Integer marks6 = 10;
        Integer marks7 = 21;
        Integer sum = marks6 + marks7;
        System.out.println("Sum of 2 wrapper class object is: " + sum);

    }
}
```

# Output:

```
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c
Primitive Variable Value: 88
Wrapper class object: 88
Wrapper class object created using autoboxing: 70
Primitive datatype using manual unboxing: 33
Primitive datatype using automatic unboxing: 45
Number is: 250
String is: 67
Sum of 2 wrapper class object is: 31
PS C:\Users\kchig\OneDrive\Desktop\java_college>
```

# Learning Objectives :

**To perform conversion between primitive data types and wrapper objects using Java wrapper classes.**

# Theory :

**Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.**

# Outcome :

**Successfully developed Java applications demonstrating the use of wrapper classes for converting primitive data types to objects and vice versa. The programs implemented boxing, unboxing, autoboxing, parsing, and conversion methods, and performed arithmetic operations and validation using wrapped values in real-world scenarios such as student marks calculation and employee payroll processing.**

# Conclusion :

**This assignment enhanced the understanding of Java wrapper classes and their importance in converting between primitive data types and objects. It demonstrated the concepts of boxing, unboxing, autoboxing, parsing (parseInt()), and conversion methods (valueOf(), toString()). The implementation also showed how wrapper objects can be used for performing arithmetic operations and input validation, thereby strengthening the understanding of object-oriented programming and data handling in Java.**

# Exercise :

## 1.Create a program to convert student marks from String format to Integer and calculate total marks.

### Program Code:

```
EXP_3_1.java > EXP_3_1 > main(String[])
public class EXP_3_1 {
    Run | Debug
    public static void main(String[] args) {
        String math_marks = "80";
        String physics_marks = "70";
        String chem_marks = "90";

        System.out.println(x: "Marks of Different Subjects(out of 100) in String Format");
        System.out.println("Math Marks: " + math_marks);
        System.out.println("Physics Marks: " + physics_marks);
        System.out.println("Chemistry Marks: " + chem_marks);

        // Converting marks in Integer format from String format.
        int marks_math = Integer.parseInt(math_marks);
        int marks_physics = Integer.parseInt(physics_marks);
        int marks_chem = Integer.parseInt(chem_marks);

        System.out.println(x: "****************************");
        System.out.println(x: "****************************");

        System.out.println(x: "Marks of Different Subjects(out of 100) after conversion in Integer Format");
        System.out.println("Math Marks: " + marks_math);
        System.out.println("Physics Marks: " + marks_physics);
        System.out.println("Chemistry Marks: " + marks_chem);

        int total = marks_math + marks_physics + marks_chem;
        System.out.println();
        System.out.println("Total marks: " + total);

    }
}
```

## Output:

```
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c:\Users\kchig\OneDrive
Marks of Different Subjects(out of 100) in String Format:
Math Marks: 80
Physics Marks: 70
Chemistry Marks: 90
****************************
****************************
Marks of Different Subjects(out of 100) after conversion in Integer Format:
Math Marks: 80
Physics Marks: 70
Chemistry Marks: 90

Total marks: 240
```

## 2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

## Problem Code:

```java
EXP_3_2.java > EXP_3_2 > main(String[])
import java.util.Scanner;

public class EXP_3_2 {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print(s: "Enter Employee ID: ");
        String empid = scanner.nextLine();
        System.out.print(s: "Enter Basic Salary: ");
        String salary = scanner.nextLine();
        System.out.print(s: "Enter Bonus: ");
        String bonus = scanner.nextLine();

        // converting String to Wrapper objects.

        Integer emp_id = Integer.valueOf(empid);
        Double basicsalary = Double.valueOf(salary);
        Double bonusamount = Double.valueOf(bonus);

        System.out.println(x: "*****Displaying Wrapper Class Objects*****");
        System.out.println("Employee ID: " + emp_id);
        System.out.println("Basic Salary: " + basicsalary);
        System.out.println("Bonus Amount: " + bonusamount);

        if (emp_id.intValue() < 0) {
            System.out.println(x: "Employee ID Cannot be Negative");
            System.exit(status: 0);
        }
        if (basicsalary.doubleValue() < 0) {
            System.out.println(x: "Salary Cannot be Negative");
            System.exit(status: 0);
        }
        if (bonusamount.doubleValue() < 0) {
            System.out.println(x: "Bonus Amount Cannot Be Negative");
            System.exit(status: 0);
        }
        Double netsalary = basicsalary + bonusamount;

        System.out.println(x: "-----Employee Payroll System-----");
        System.out.println("Employee ID: " + emp_id);
        System.out.println("Basic Salary: " + basicsalary);
        System.out.println("Bonus Amount: " + bonusamount);
        System.out.println("Net Salary: " + netsalary);

        scanner.close();
```

**Output:**

```
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c:\Users\kchig\C
Enter Employee ID: -1
Enter Basic Salary: 50000
Enter Bonus: 1000
*****Displaying Wrapper Class Objects*****
Employee ID: -1
Basic Salary: 50000.0
Bonus Amount: 1000.0
Employee ID Cannot be Negative
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c:\Users\kchig\C
Enter Employee ID: 101
Enter Basic Salary: -78
Enter Bonus: 80
*****Displaying Wrapper Class Objects*****
Employee ID: 101
Basic Salary: -78.0
Bonus Amount: 80.0
Salary Cannot be Negative
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "c:\Users\kchig\C
Enter Employee ID: 101
Enter Basic Salary: 10000
Enter Bonus: -7
*****Displaying Wrapper Class Objects*****
Employee ID: 101
Basic Salary: 10000.0
Bonus Amount: -7.0
Bonus Amount Cannot Be Negative
```

```
PS C:\Users\kchig\OneDrive\Desktop\java_college> cd "
Enter Employee ID: 101
Enter Basic Salary: 10000
Enter Bonus: 1000
*****Displaying Wrapper Class Objects*****
Employee ID: 101
Basic Salary: 10000.0
Bonus Amount: 1000.0
-----Employee Payroll System-----
Employee ID: 101
Basic Salary: 10000.0
Bonus Amount: 1000.0
Net Salary: 11000.0
PS C:\Users\kchig\OneDrive\Desktop\java_college>
```